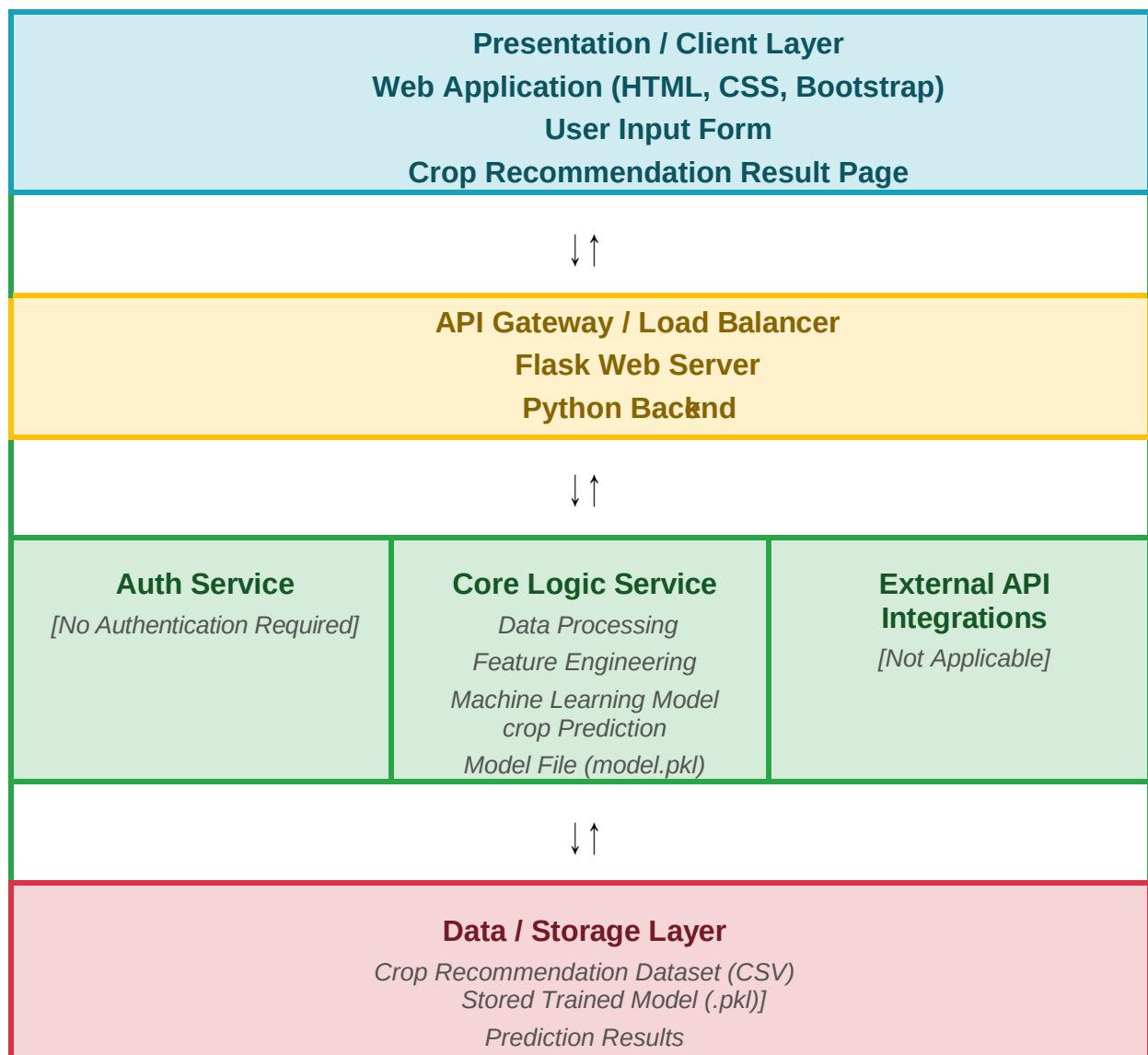


Solution Architecture

Date	02 July 2026
Team ID	XXXXXX
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Solution Architecture Diagram:

Provide a high-level visual representation of your solution's architecture. Use the structural layout below as a template to map out your Presentation Layer, Application Layer, and Data Layer. Fill in the specific technologies and components for your project directly into the boxes to create your architectural diagram.



Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
<i>[Presentation Layer]</i>	<i>Provides interface for user to enter soil and environmental details to view prediction result</i>	<i>HTML, CSS, BOOTSTRAP</i>
<i>[API Gateway]</i>	<i>Receives user input and sends data to machine learning model for prediction.</i>	<i>Flask, Python</i>
<i>[Microservices]</i>	<i>Processes input data, performs preprocessing and predicts the most suitable crop</i>	<i>Scikit-learn, Pandas, NumPy</i>
<i>[Database]</i>	<i>Stores crop dataset and trained machine learning model.</i>	<i>CSV Files, Pickle Model</i>